

MULTI-AGENT AI FOR GAME DEVELOPMENT

Emberkeep

Game Design Document — v0.5

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Executive Summary

Emberkeep is a 1–4 player co-op roguelike dungeon crawler set in the Undervault, a kingdom whose lamp-halls went dark floor by floor a thousand years ago when something at the bottom woke. Each run, a hero descends to rally a growing retinue: every class recruits, awakens, bonds, or kindles allies that fight alongside them, escalating from a handful of units to screen-filling hordes rendered in a 2-bit palette built specifically to stay legible at that scale.

This is a roguelike first. The run is the primary loop, and **retinue growth is the progression axis and the run's reward**. The RTS command layer expresses that power moment-to-moment; the persistent world adds seasoning between runs — both serve the spine.

Loss is hero death — the run ends immediately. **Per-run win** is clearing the floor's boss and reaching the exit. **Meta-win** is longer-term: power resets every run, but the decisions you make persist — they write to a small set of world flags that the next run's procedural generation reads, so across many runs the Hollow

Crown's grip on the Undervault visibly weakens. Those decisions are protected by a firm rule: every world-scarring choice goes to a **party vote**, and a passed vote writes to every present player's own save alike.

The project is built by a small multi-agent AI development pipeline (Section 3): specialized director agents own the lore, the readability rules, and the numbers; MCP tool servers let those agents edit the live Unreal Engine 5.8 project and generate pixel art directly; and a **review board** of six agents adversarially audits the design canon itself — the reason every world-scarring choice requires the whole party's agreement is that this board exists to find exactly that kind of gap.

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Game Mechanics

PLAYER-FACING ACTIONS AND LOOP

- 1 Pick your class.** Four working roster slots, each a genuinely different game: **Vanguard** (melee anchor; retinue of liberated soldiers who hold ranks and form shield walls), **Relickeeper** (fortifier; awakens slow, durable stone guardians and inscribes ward zones), **Pathfinder** (ranged skirmisher; a small named hunting pack of bonded beasts and scouts), **Lampbearer** (healer/vision; kindles a drifting constellation of captured souls that heal instead of fight). Every class starts locked to its own retinue identity.
- 2 Descend and grow your army.** Enter a procedurally generated floor and fight through combat encounters. Retinue size starts at 3–10 units and can reach the hundreds by late run — grown by rescuing prisoners, awakening dormant guardians, bonding wounded creatures, or kindling lost souls, depending on class. Growth sites scale with party size — a floor guarantees at least one per player, keeping a full 4-player party fed and a solo player's pace comfortable. Feed your army and it stays strong: every unit draws on a shared supply, and a unit that outruns it visibly **degrades** — dimmer, weaker — until you catch back up. The army is governed by upkeep — a living economy that breathes with the fight.

- 3 **Command with broad stances.** One button/wheel issues a single order to the whole retinue: **Follow** (default formation), **Charge**, **Hold**, **Rally**. Each class reflavors these same four verbs — the Vanguard's Hold is a literal Shield Wall that blocks enemy pathing; the Pathfinder's Charge, *Loose the Pack*, sends the pack hunting off-screen. The **leash rule** keeps the army with you: any unit that strays too far from the hero — including one on Hold — snaps back to Follow, so anchoring a chokepoint always means staying in the fight beside your troops.
- 4 **Face a personal decision.** Each floor includes at least one decision event targeted at a specific player: a fork (free the caged prisoners, raising the alarm, vs. loot the vault quietly), a sacrifice (trade retinue lives or hero HP for a power spike), or a moral choice with consequences that follow you past this run. A Vanguard's permadeath Veterans and a Pathfinder's pack member who's merely "found again, changed" carry very different weight — so sacrifice offers price what you'd actually feel losing, giving every class's choice equal weight. Hoarding carries its own cost: a Lampbearer who keeps gathered souls burning watches them dim and grow restless — **unrest**, the mounting toll of temptation.
- 5 **Clear the floor — the per-run win condition.** Beat the floor's boss and reach the exit. Three floors escalate in density toward a final confrontation.
- 6 **Die, and the run ends — the loss condition.** Hero death ends the run immediately and finally.
- 7 **The world remembers — the meta-win condition.** When the run ends (win or death), all hero and retinue power resets to zero for the next run — but the world flags your decisions wrote persist: a site you saved might become a safe room next time; an NPC you spared might reappear as a recruitable ally. Anything that would scar the world for the whole party goes to a **party vote** (simple majority; ties and absent players default to the status quo), and a passed vote writes to **every present player's own world** alike — every world-altering choice reflects the whole party's will.

The game is built by a small team of specialized development agents plus two MCP tool servers that give those agents direct hands on the engine and the art pipeline. Each canon file (lore, art specs, systems data) has exactly one agent with write access, and every other agent treats it as read-only — canon conflicts are caught by that structure itself. A seventh role, the review board, audits the canon.

AGENT / TOOL	DEVELOPMENT ROLE	WHAT THE PLAYER EXPERIENCES
narrative-director	Writes the world's lore, faction fiction, decision-event text, and NPC dialogue a player reads at event sites, then hands off an art brief for anything visual.	The prisoner you free at a rescue site has a name, a reason to fear the Legion, and a line that changes depending on whether you saved or ignored her last run.
pixel-art-director	Defines the 2-bit silhouette and value rules per faction and reviews sprite-sheet layouts for readability, producing palette tables and written specs.	In a 200-unit battle you can tell your Liberated ranks from Legion soldiers from the Quiet's snuffers at a glance, by shape and value alone.
gameplay-director	Owens the stat blocks, scaling curve, and encounter budget in SYSTEMS.md / docs/data — the numbers behind "a handful of peasants" becoming a screen-filling army.	Floor 1 feels survivable with three militiamen; floor 3's boss room throws a wave your army can actually break, because someone tuned the curve.
unreal-mcp (tool)	Exposes the live UE 5.8 editor — actors, Blueprints, Niagara systems — over HTTP so an agent can build and wire gameplay directly in-editor instead of just describing it in a document.	The stance wheel, the leash-break-to-Follow behavior, and the swarms you fight in the Spike 1 prototype were all built through this channel.
pixellab (tool)	An MCP-driven generation service that produces character sprites, portraits, tiles, and animations to the pixel-art-director's	The Vanguard's portrait and the Liberated militia sprite are generated here and only promoted to the game's

AGENT / TOOL	DEVELOPMENT ROLE	WHAT THE PLAYER EXPERIENCES
	spec, with every result reviewed before it ships.	actual art once a human confirms they read correctly at scale.
Orchestrator	Reads the design canon, briefs each director with only the section it owns, and integrates their output back into the docs and the live UE project.	Keeps independent agents from drifting out of sync — a stance-rule change gets propagated everywhere it's referenced instead of quietly rotting in one file.
GDD Review Board (<code>gdd-review-kit</code>)	Six isolated-context reviewer agents (systems-designer, narrative-critic, player-psychologist, feasibility-lead, adversarial-qa, business-analyst) independently critique the design canon, cross-examine each other's findings in a second round, then a moderator synthesizes a ranked, ratifiable edit plan.	The reason every shared NPC's fate and every faction's standing reflects the whole party's agreement is that an adversarial-QA pass on this exact document found the gap — "the party owns the scar" is a rule in the game because an agent argued for it.



Technical Strategy

ROLES, TOKEN BUDGET, CONSTRAINTS

Model-tier assignment — grounded in measured usage to date. Directors run mostly on **Claude Fable 5** (open-ended lore/spec work, the historical workhorse) and **Claude Sonnet 5** (the current default for structured/technical passes). **Claude Opus 4.8** is reserved for the highest-stakes passes — cross-file canon integration, and the entire review board, which runs entirely on Opus 4.8 (104 turns, measured): adversarial critique of your own design is exactly the high-stakes, low-volume case that tier exists for.

METRIC	MEASURED VALUE	NOTES
Fresh tokens to date (main dev project)	~22.25M	across 3,731 assistant turns, 2026-07-09–07-19

METRIC	MEASURED VALUE	NOTES
Review board pass	~850k fresh	104 turns, one full five-round cycle, entirely Opus 4.8 — measured from <code>gdd-review-kit</code> 's own session logs, a separate project
Cache-read tokens to date	~552.65M (main) + ~6.36M (review)	replayed context, heavily discounted
Semester total (projected)	~55–95M fresh tokens + ~850k per review cycle	a review pass is a known, cheap, recurring line item — roughly 1% of the semester budget each time it runs

Named constraints.

- **Single shared UE 5.8 editor instance (the binding constraint).** All in-engine work goes through one live editor process running the `unreal-mcp` server on port 8000. UE-touching work is serialized — one agent, one call, at a time — while directors continue writing docs/specs in parallel.
- **Entity-count spike gate — the semester's real go/no-go.** The core hook — hundreds-to-thousands of Mass Entity units at 60fps under 4-player replication — is a hard, enforced gate: content and class production begin once Spike 2 passes at the **full 4-client target load**. A gate failure invalidates the concept, and the evaluated fallbacks are reduced entity counts, single-player-only mass, or a fake-crowd hybrid.
- **Pre-resourcing posture.** Team size, budget, and ship window are explicitly TBD. The v1 scope tables describe intent; costing begins once a team exists.

Semester-realistic scope. 1 class (Vanguard), 1 biome (the Highgates), 3 floors, 1 boss, 2 decision events, 2–4 player replication, and a stubbed world-flag write, built against a governed retinue economy and a party-vote multiplayer model. The Gatecamp hub stays deferred to the next biome.

